

Status Report for EMBL Hamburg

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19th May 2025

MXCuBE-ISPyB Meeting
DESY, Hamburg



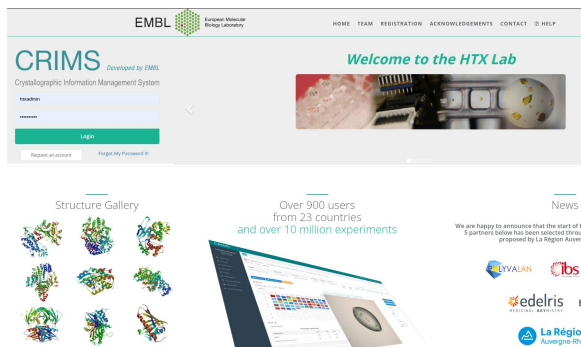
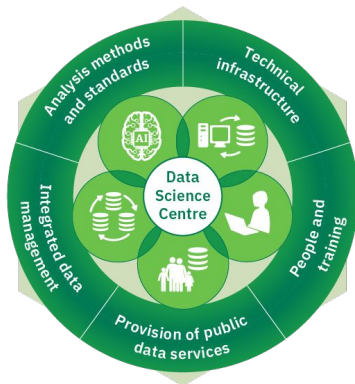
Introduction to self

- Masters in Bioinformatics and Applied Biotechnology, India
- Formerly, a Bioinformatics Engineer in India
- Structural Biology Data Platform Developer

EMBL Heidelberg
Data Science Center

EMBL Grenoble
HTX Lab

EMBL Hamburg
Schneider Group



ICAT/DRAC Progress Update

EMBL Grenoble

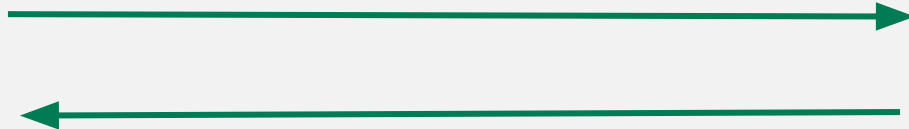
CRIMS-ICAT Communication

EMBL Hamburg

ICAT/DRAC Installation and deployment

ICAT/DRAC Progress Update - EMBL Grenoble

Create Shipments: Information about Containers, samples, experiment and processing plans



Download Datasets - Cronjob, checks for dataset availability for every shipment sent

Data flow

CRIMS *Developed by EMBL*

Crystallographic Information Management System

Sample information
Experiment workflow



Data Collection



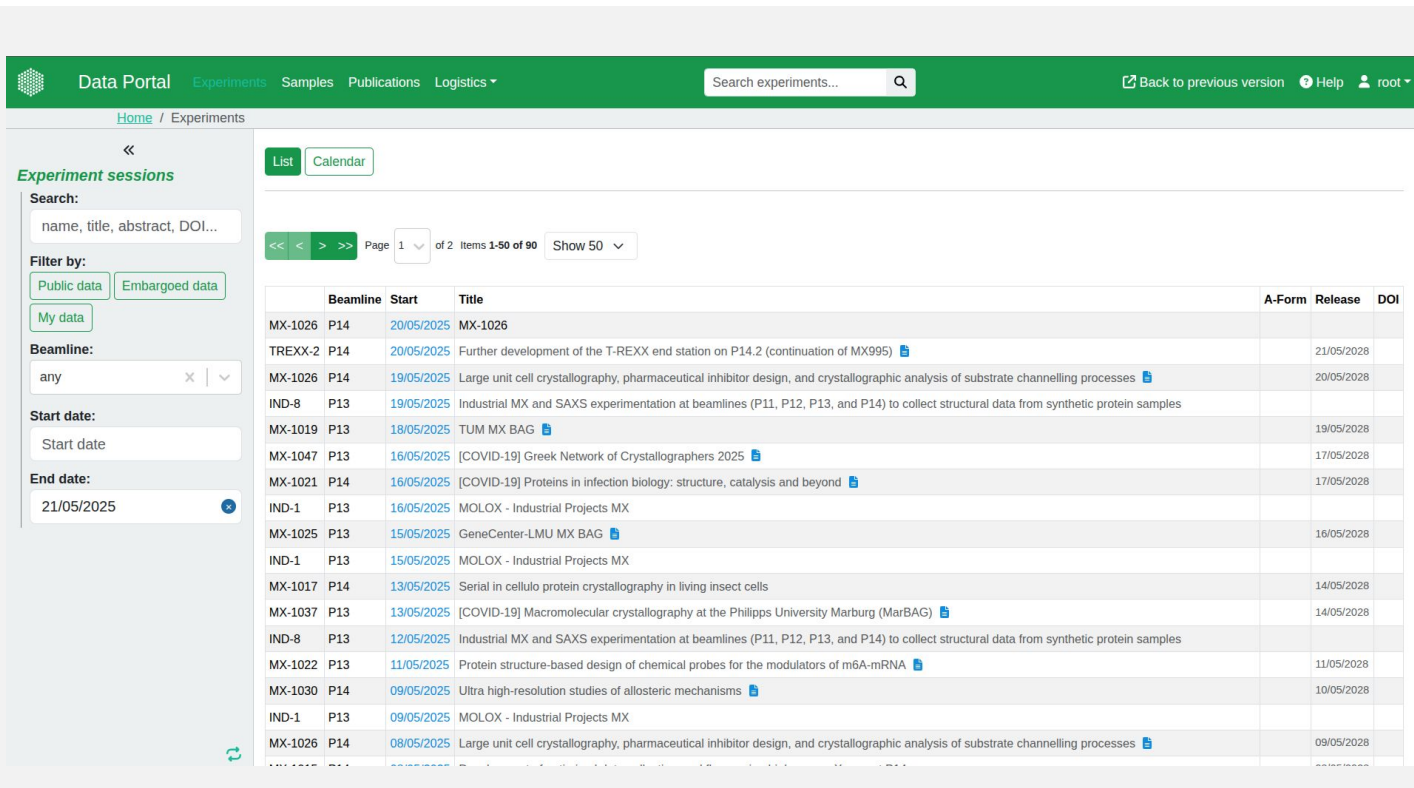
Sample information
Experiment workflow



Data Collection



ICAT/DRAC Progress Update - EMBL Hamburg



The screenshot displays the EMBL Data Portal interface for the 'Experiments' section. The top navigation bar includes links for Data Portal, Experiments, Samples, Publications, and Logistics. A search bar is present with the text 'Search experiments...'. Below the navigation bar, the 'Experiments' page is shown with a 'List' button and a 'Calendar' button. A search bar is labeled 'Search: name, title, abstract, DOI...'. A filter section on the left includes 'Filter by:' with buttons for 'Public data', 'Embargoed data', and 'My data'. Below this, the 'Beamline:' filter is set to 'any'. The 'Start date:' filter is set to 'Start date'. The 'End date:' filter is set to '21/05/2025'. The main content area shows a table of experiment sessions with columns for Beamline, Start, Title, A-Form, Release, and DOI. The table lists various experiments, including MX-1026, TREXX-2, MX-1026, IND-8, MX-1019, MX-1047, MX-1021, IND-1, MX-1025, IND-1, MX-1017, MX-1037, IND-8, MX-1022, MX-1030, IND-1, and MX-1026.

	Beamline	Start	Title	A-Form	Release	DOI
MX-1026	P14	20/05/2025	MX-1026			
TREXX-2	P14	20/05/2025	Further development of the T-REXX end station on P14.2 (continuation of MX995)		21/05/2028	
MX-1026	P14	19/05/2025	Large unit cell crystallography, pharmaceutical inhibitor design, and crystallographic analysis of substrate channelling processes		20/05/2028	
IND-8	P13	19/05/2025	Industrial MX and SAXS experimentation at beamlines (P11, P12, P13, and P14) to collect structural data from synthetic protein samples			
MX-1019	P13	18/05/2025	TUM MX BAG		19/05/2028	
MX-1047	P13	16/05/2025	[COVID-19] Greek Network of Crystallographers 2025		17/05/2028	
MX-1021	P14	16/05/2025	[COVID-19] Proteins in infection biology: structure, catalysis and beyond		17/05/2028	
IND-1	P13	16/05/2025	MOLOX - Industrial Projects MX			
MX-1025	P13	15/05/2025	GeneCenter-LMU MX BAG		16/05/2028	
IND-1	P13	15/05/2025	MOLOX - Industrial Projects MX			
MX-1017	P14	13/05/2025	Serial in cellulose protein crystallography in living insect cells		14/05/2028	
MX-1037	P13	13/05/2025	[COVID-19] Macromolecular crystallography at the Philipps University Marburg (MarBAG)		14/05/2028	
IND-8	P13	12/05/2025	Industrial MX and SAXS experimentation at beamlines (P11, P12, P13, and P14) to collect structural data from synthetic protein samples			
MX-1022	P13	11/05/2025	Protein structure-based design of chemical probes for the modulators of m6A-mRNA		11/05/2028	
MX-1030	P14	09/05/2025	Ultra high-resolution studies of allosteric mechanisms		10/05/2028	
IND-1	P13	09/05/2025	MOLOX - Industrial Projects MX			
MX-1026	P14	08/05/2025	Large unit cell crystallography, pharmaceutical inhibitor design, and crystallographic analysis of substrate channelling processes		09/05/2028	

- Installed of containerized DRAC Application on a RHEL VM
- Established connection between SMIS and DRAC
- Proposal information ingested from SMIS to DRAC

ICAT/DRAC Progress Update - EMBL Hamburg

Enabled the MX Viewer
for Datasets

Data Portal Experiments Samples Publications Logistics ▾

Search experiments... 🔍

Back to previous version ⓘ Help ⓘ InvestigationUserFullName ▾

Home / TEST-0001 P14 01/01/2025 - 04/01/2026 / Datasets

Experiment session

Statistics

Datasets 37

Logbook

Prepare

Datasets

View as:
List Summary

Sample:
Type sample name ▾

Protein:
☐ agro1
☐ ta

Collection steps

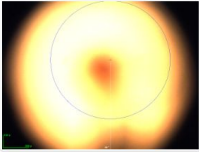
Show snapshots for:

- ☒ Data collection
- ☒ Automesh
- ☒ Mesh
- ☒ Line
- ☒ Characterisation

29/04/2025 20:51 MY_OTHER_SAMPLE_TODAY

29/04/2025 20:51:31 MXPressA: X-centre, eEDNA + dc on id30a1

Mesh ⓘ 🔍



29/04/2025 20:51 MY_SAMPLE_TODAY

29/04/2025 20:51:31 MXPressA: X-centre, eEDNA + dc on id30a1

Mesh ⓘ 🔍

Snapshot not found

16/04/2025 15:30 to 07/05/2025 16:16 my_sample

07/05/2025 16:16:41 mesh

29/04/2025 20:51:31 MXPressA: X-centre, eEDNA + dc on id30a1

Mesh ⓘ 🔍 Mesh ⓘ 🔍

ICAT/DRAC Progress Update - EMBL Hamburg

Enabled the MX
Viewer for Datasets

Data Portal Experiments Samples Publications Logistics Search experiments... Back to previous version Help InvestigationUserFullName

Home / TEST-000 / 01/01/2025 - 04/01/2026 / Datasets

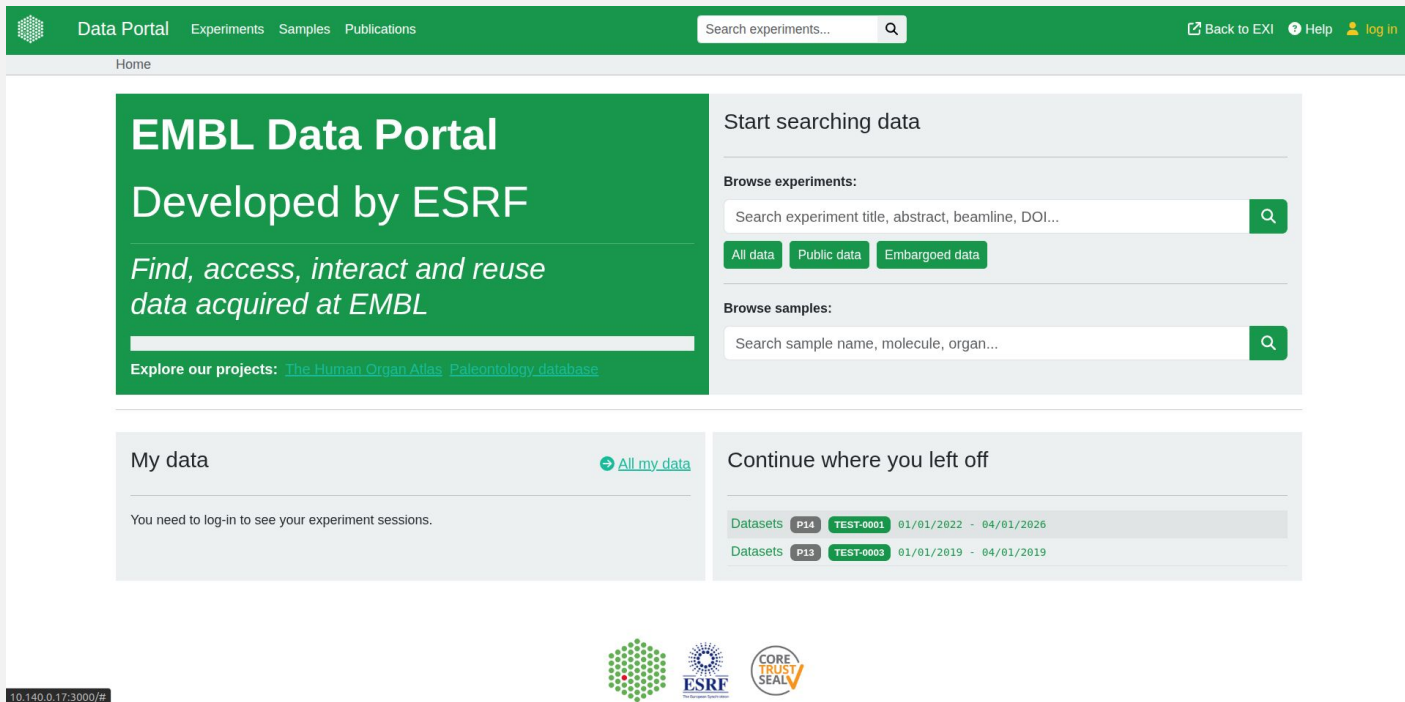
Mesh Metadata Sample Files 2

Acquisition		Beam		Sample	
Dataset name	mesh	Res. (corner)		Protein	agro1
Start time	29/04/2025 20:51:31	Wavelength	0.97	Sample	MY_OTHER_SAMPLE_TODAY
Position		Axis range		Parcel	
Workflow	MXPressA	Axis start (end)	--	Container name	cps3756
Prefix	mesh-agro1-G9_3_1_2_%04d.cbf	Exposure Time	0.02	Container type	UNIPUCK
Run #		Flux start	7.76e+11 ph/s	Position in container	10
# Images	900	Flux end	7.67e+11 ph/s	Position in changer	2-3
Transmission	100.0	X Beam	206.07 mm	Beamline location	P14
		Y Beam	224.49 mm		

Optics		Synchrotron		Detector	
Focussing optics	NA	Name		Type	
Mono type	NA	Filling mode		Model	
Beam size X	NA µm	Current		Manufacturer	
Beam size Y	NA µm			Pixel size horizontal	0.172
Beam shape	ELLIPTICAL			Pixel size vertical	0.172
Beam divergence Hor	NA µrad			Distance	
Beam divergence Vert	NA µrad				
Polarization	NA				

Datasets
View as: List Summary
Sample: Type sample name
Protein: agro1 ta
Collection steps: Show snapshots for: Data collection Automesh Mesh Line Characterization

ICAT/DRAC Progress Update - EMBL Hamburg

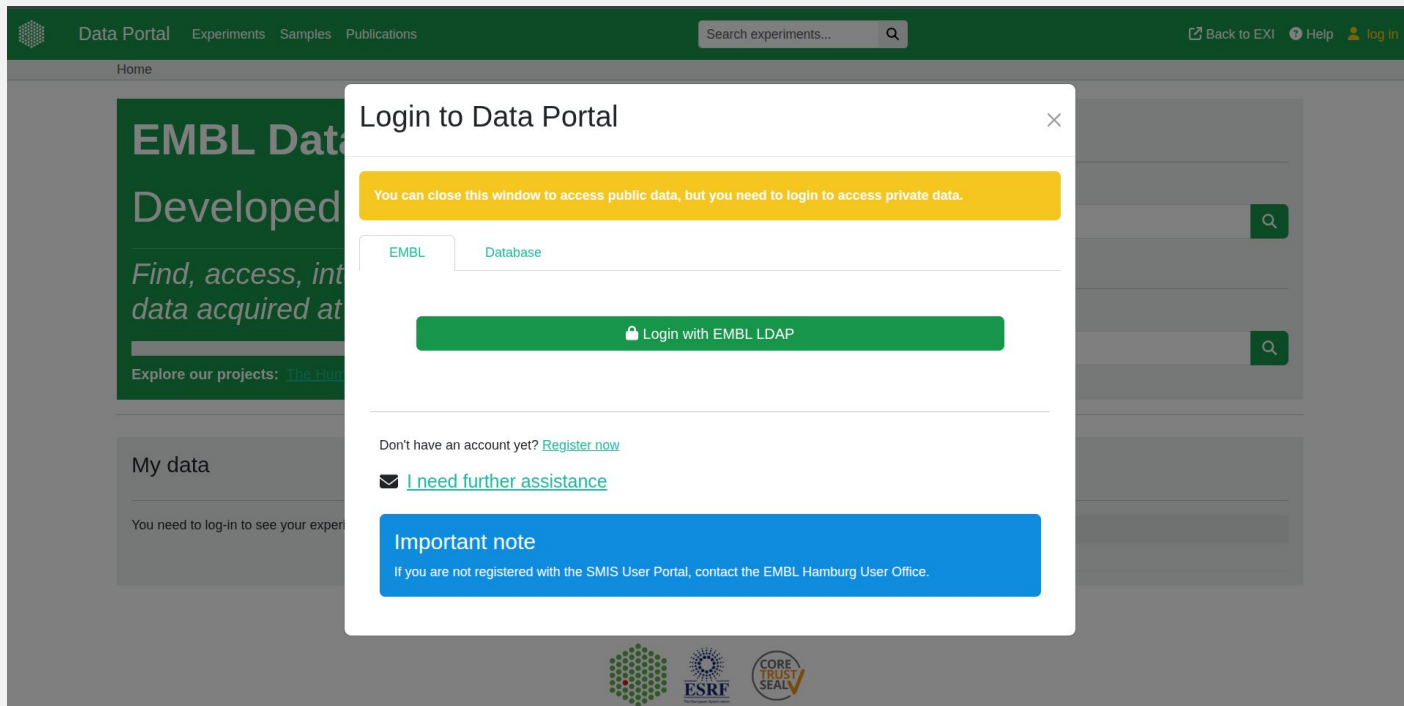


The screenshot displays the EMBL Data Portal interface. At the top is a green navigation bar with the EMBL logo, links for 'Data Portal', 'Experiments', 'Samples', and 'Publications', a search bar for experiments, and links for 'Back to EXI', 'Help', and 'log in'. Below the navigation bar is a 'Home' breadcrumb. The main content area is divided into several sections. On the left, a large green box contains the text 'EMBL Data Portal Developed by ESRF' and 'Find, access, interact and reuse data acquired at EMBL', followed by links to 'The Human Organ Atlas' and 'Paleontology database'. To the right of this is a 'Start searching data' section with 'Browse experiments' and 'Browse samples' search bars, each with a search button and filter buttons for 'All data', 'Public data', and 'Embargoed data'. Below these is a 'My data' section with a link to 'All my data' and a message: 'You need to log-in to see your experiment sessions.' To the right of 'My data' is a 'Continue where you left off' section showing a list of datasets with IDs like P14, TEST-0001, and P13, TEST-0003, along with their respective date ranges. At the bottom of the page are logos for ESRF, CORE TRUST SEAL, and a hexagonal grid logo.

10.140.0.17:3000/#

- UI Styles Modifications to fit with EMBL theme

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- UI Styles Modifications to fit with EMBL theme

Next Steps



- **CRIMS-DRAC**
 - Authentication
 - Automation of dataset retrieval
- **DRAC Hamburg**
 - Enable LDAP authentication
 - Establish MXCuBE-DRAC communication

Acknowledgements

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HTX Lab Members

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